Mark	2/11

Team name:	B9		
Homework number:	Homework 02		
Due date:	30/09/2025		
Contribution:	**No**	**Partial**	**Full**
Pica Mirko			✅
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Notes:			

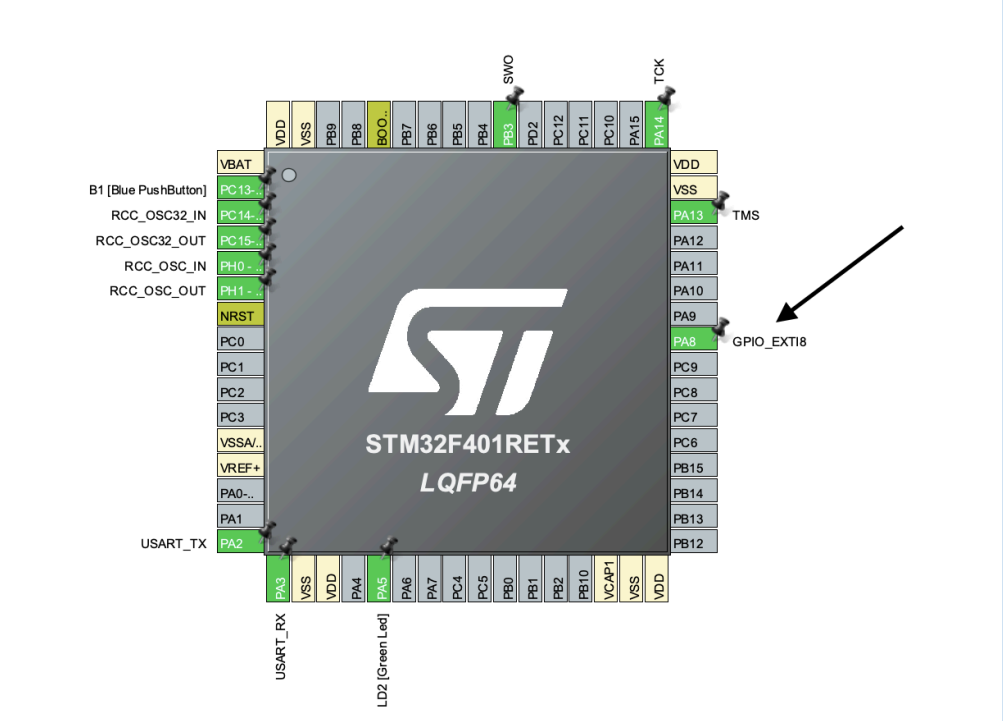
Project name	Prepare the basics for the project “Play a song”		
Not done	**Partially done (major problems)**	**Partially done (minor problems)**	**Completed**
			✅

We successfully completed the project by doing the following.

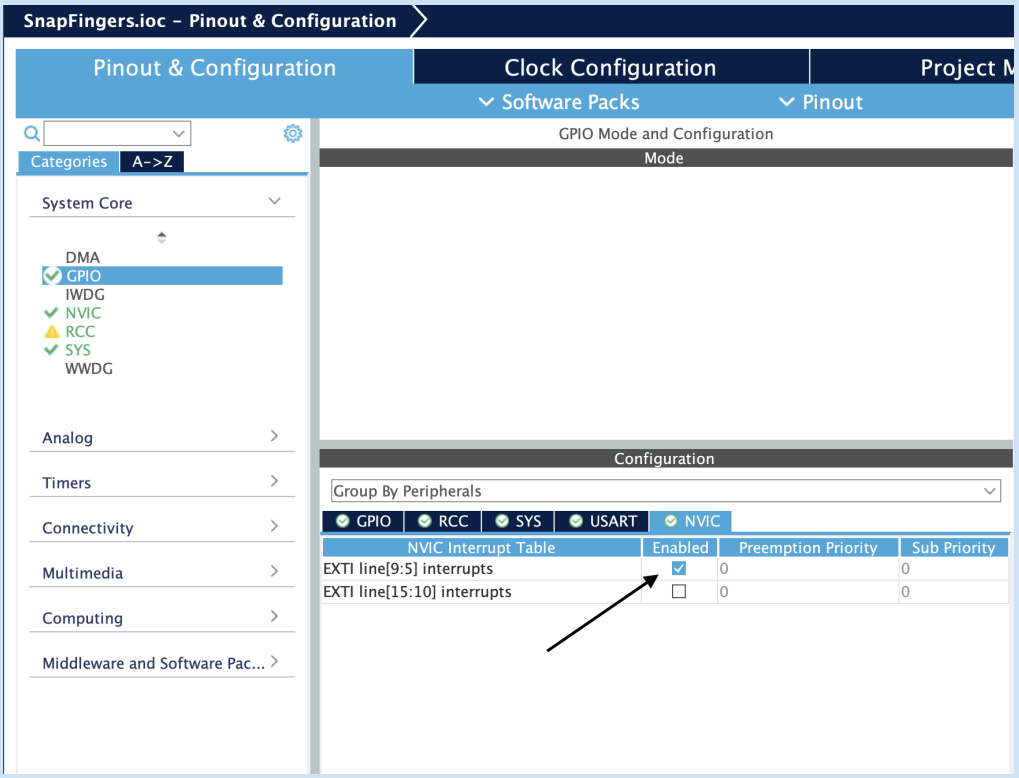
Part 1a. Modify the status (switch on / off) of the NUCLEO green LED, every time you snap your fingers. (Use the pin connected to the microphone as an External Interrupt)

In the “Hands-on Lab Schematics.pdf”, we found that the pin PA8 is connected to our microphone: SND_IN.

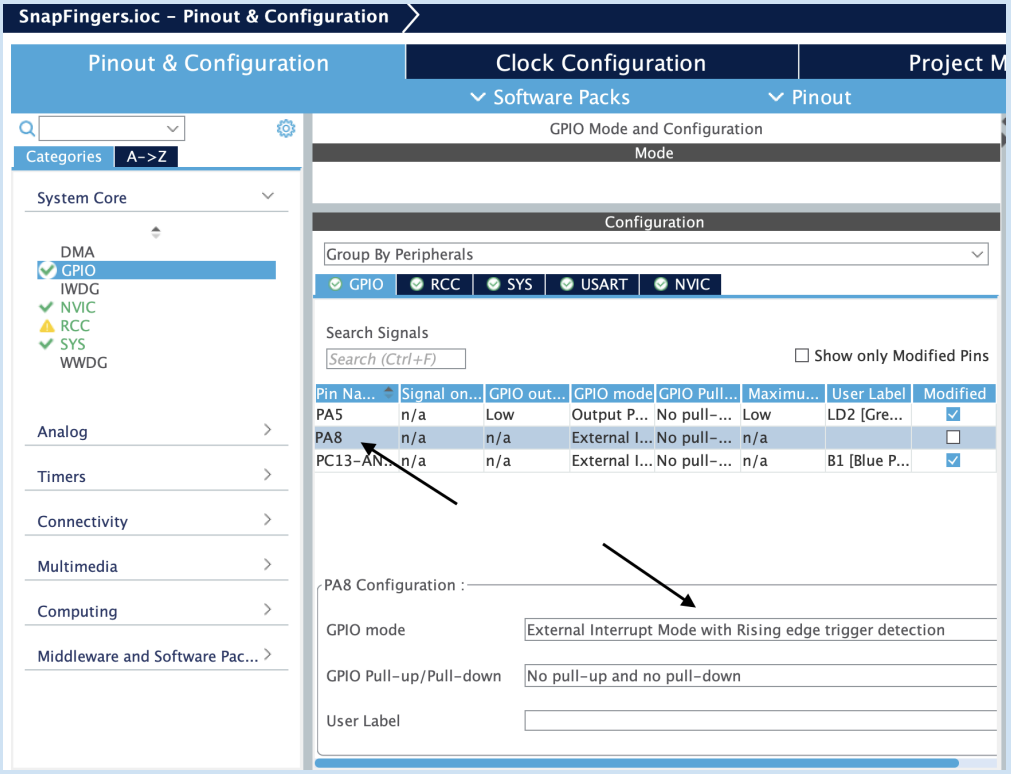
In the IDE we configure the pin PA8 as GPIO_EXTI8:



We enable the EXTI interrupt as shown below:



We configure the pin PA8 as shown below:



This way we generate an interrupt every time there is a rising edge, so that every time we make a sound to the microphone, the state of the LED changes.

Finally we go to the “main.c” and add the code as shown below:

```
/* USER CODE BEGIN 0 */
void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin) {
    if (GPIO_Pin == GPIO_PIN_8) {
        HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);
    }
}
```

Which means that every time the interrupt is triggered and the request is from the microphone pin, the state of the LED changes. We have tested it and it works, but we observed a problem.

When we snap or clap one time, the LED blinks (turn off and turn on) many times. We noticed that the input signal (from mic) is not clean, it bounces, meaning the microcontroller can see many fast pulses instead of just one. To avoid this, we add a debounce period:

After detecting one snap, we ignore any new snaps for 200 ms. That way, only one action happens per snap.

We defined the following variables:

```
volatile uint32_t last_snap_tick = 0;// Stores last snap time in ms
#define SNAP_DEBOUNCE_MS 200 // 200ms debounce
```

And updated the interrupt handler in the following way:

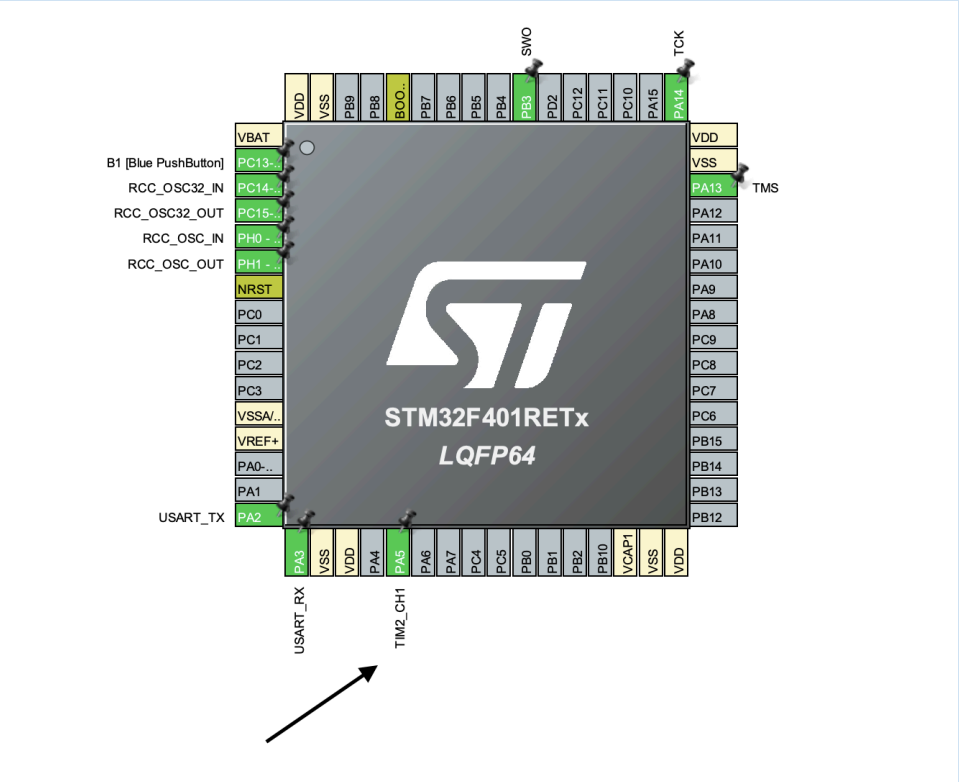
```c
void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin) {
    if (GPIO_Pin == GPIO_PIN_8){
        uint32_t current_time = HAL_GetTick();
        if ((current_time - last_snap_tick) > SNAP_DEBOUNCE_MS){
            HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);
            last_snap_tick = current_time;
        }
    }

}
```

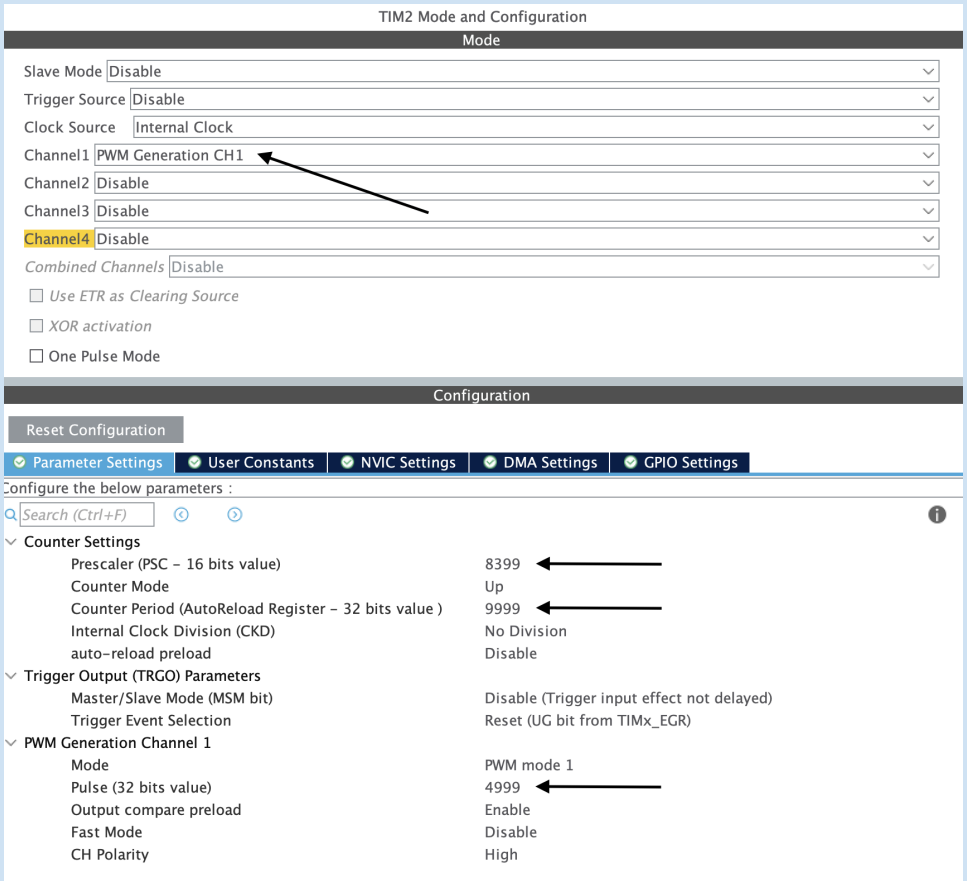
We are curious if there is a way to manage the sensibility of the microphone.

Part 1b. Make the NUCLEO green LED blink at a 1 Hz rate using PWM generation on the corresponding channel.

In the IDE, we have configured the pin PA5 (green led) to TIM2_CH1:



We configured the TIM2 as shown below:



We set Prescaler at 8399, Auto Reload Register to 9999 and Pulse to 4999 which gets us to 1 Hz of frequency and 50% duty cycle, following the formula presented in class.

PWM frequency: $$f_{PWM} = \frac{f_{TIM}}{(ARR+1)\cdot(PSC+1)}$$

PWM Duty Cycle: $$DC = \frac{CCRx+1}{ARR+1}$$

Finally in the “main.c”, we put the code as shown below to start the timer:

```
/* USER CODE BEGIN 2 */
HAL_TIM_PWM_Start(&htim2, TIM_CHANNEL_1);
/* USER CODE END 2 */
```

We have tested it and it works.

Professor comments:	